FORGE:Fine-grained Multimodal Evaluation for Manufacturing Scenarios

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Research Questions

- Can MLLMs understand, explain, and execute decisions for tasks inherently characteristic of the manufacturing domain?
- What are the primary bottlenecks limiting MLLM performance in fine-grained manufacturing scenarios?
- How can domain-specific fine-tuning improve MLLM performance on manufacturing tasks?

Core Idea

Existing MLLM benchmarks for manufacturing lack fine-grained domain semantics (e.g., exact model numbers like M10 vs M20 screws) that are critical for real manufacturing scenarios. The paper creates FORGE, a benchmark with multimodal data (2D images + 3D point clouds) annotated with precise manufacturing details, and evaluates 18 MLLMs on three tasks: workpiece verification, surface inspection, and assembly verification. Surprisingly, visual grounding isn't the main bottleneck - insufficient domain-specific knowledge is. The key insight is that 3D point clouds are converted to multi-view 2D projections since MLLMs lack native 3D encoders, and fine-tuning small models on their structured annotations yields substantial improvements.

Key Figure

Table 3: Main benchmark results (accuracy %) across three manufacturing tasks. **Bold orange** / underlined blue / light purple indicate best / second / third within each model category (open- or closed-source) per row. **Rand.** = weighted random-guess baseline.

Task	Mod.	Setting	Open-source Models										Closed-source Models								Random
			Gemma-3-27B	InternVL3-78B	Llama-4-MAV	Mistral-3-14B	Mistral-3-8B	Mistral-3-Large	Qwen3-VL-235B	Qwen3-VL-8B	Kimi-K2.5	GLM-4.6V	Claude-4.5-Opus	Gemini-2.5-Flash	Gemini-3-Flash	GPT-5.2	GPT-5	GPT-5 Mini	O3	Seed-1.6	
WorkVeri	3V	Zero Shot	27.62	32.59	36.36	32.46	30.97	34.78	52.36	42.01	50.00	47.42	39.19	56.05	69.56	61.29	61.05	37.98	58.22	54.44	25.0
		Ref-Cond	23.59	24.90	20.36	25.00	18.26	20.93	35.67	27.20	36.76	44.29	27.88	43.55	65.25	28.34	30.02	33.94	29.55	26.87	25.0
		ICD	28.83	30.16	35.69	28.54	28.86	37.53	37.86	26.32	60.56	44.90	55.04	53.85	67.34	61.69	56.68	30.71	57.91	46.45	25.0
	Image	Zero Shot	25.94	32.59	38.80	33.56	30.61	25.78	64.08	35.41	66.75	50.44	59.42	55.78	72.22	72.50	74.72	73.78	76.18	67.04	25.0
		Ref-Cond	30.51	53.88	40.65	20.22	24.28	25.50	58.01	24.04	69.67	50.33	56.22	51.66	76.27	55.48	64.22	72.73	68.32	49.66	25.0
		ICD	34.59	65.26	54.19	27.20	32.07	31.18	66.44	25.28	68.86	50.00	60.98	54.55	82.26	79.87	85.23	77.01	82.49	70.29	25.0
AssyVeri	3V	Zero Shot	27.18	34.30	38.19	28.16	24.60	31.64	42.33	39.16	78.26	39.60	42.07	44.98	47.25	54.22	53.90	49.51	52.38	41.69	32.8
		Ref-Cond	32.04	22.01	30.10	19.09	18.45	29.54	31.01	33.77	57.14	37.29	50.81	40.13	46.28	56.96	30.74	29.55	36.46	26.71	32.8
		ICD	32.04	28.80	37.54	26.06	29.55	27.03	43.26	38.96	53.54	37.86	45.45	44.01	47.57	54.07	51.95	45.31	52.46	34.30	32.8
	Image	Zero Shot	28.47	33.14	36.42	30.15	29.64	29.12	36.97	31.89	33.83	35.46	52.10	39.56	58.11	48.18	50.06	43.69	48.93	39.86	29.5
		Ref-Cond	31.70	36.40	32.12	32.12	26.84	23.74	40.82	28.50	52.08	37.28	56.36	32.32	70.61	56.28	49.77	51.35	60.16	42.49	29.5
		ICD	32.94	42.46	46.13	29.46	30.65	28.67	50.43	30.32	47.72	45.33	62.92	46.89	71.50	63.99	60.94	52.29	62.34	48.52	29.5
SurfInsp	3V	Zero Shot	21.75	19.19	27.02	28.30	24.26	19.83	19.16	19.40	13.19	23.45	8.72	17.23	18.51	16.63	22.01	17.02	21.11	22.60	20.0
		Ref-Cond	23.88	21.63	24.09	27.72	27.08	19.79	18.74	21.28	16.81	23.83	7.66	26.38	29.57	21.91	35.74	33.40	36.25	36.17	20.0
		ICD	33.19	25.74	39.15	33.19	38.94	26.65	32.15	25.75	30.06	38.38	44.35	38.09	47.12	31.70	38.30	36.25	40.00	42.31	20.0

performance on SurfInsp further indicates significant room for improvement in current MLLMs' understanding of physical details and microscopic features of manufacturing workpieces.

B.Limited comprehension of domain knowledge is the bottleneck for current MLLMs. By comparing evaluation results across Zero-Shot, Ref-Cond, and ICD-based settings based on image modality, we observed that in WorkVeri and AssyVeri, simple Ref-Cond strategies did not consistently yield performance gains and, in several instances, even led to degradation. Conversely, ICD methods incorporating complete reasoning demonstrations achieved universal improvements over Zero-Shot baselines. This suggests that for complex manufacturing tasks, MLLMs do not lack simple sample references, but rather a deep understanding of task logic and reasoning paths, a gap that ICD bridges but Ref-Cond cannot. Notably, Section 4.5 shows that visual grounding is not the bottleneck: MLLMs can identify these workpieces. Error case 1 in the Appendix C.1 also corroborates that MLLMs reveal a disconnect between perception and comprehension.

C. Given limited perceptual understanding of 3D spatial contexts, the introduction of additional examples hinders comprehension of MLLMs. Based on Table 3, for three-view modality, MLLMs achieved optimal performance under the Zero-Shot setting, whereas performance declined following the introduction of Ref-Cond and ICD. Given that the fundamental distinction between the three-view and image modalities lies in their multi-angular spatial representations, this counterintuitive phenomenon suggests that the contextual examples provided in Ref-Cond and ICD do not provide effective guidance. Instead, they induced spatial confusion within MLLMs, thereby impeding comprehension of manufacturing domain knowledge. Consequently, compared with model-number-level recognition, which relies less on spatial visual features, workpiece-level recognition, which is highly dependent on visual perception, is more severely affected by Ref-Cond and ICD. Furthermore, the experimental results regarding point cloud inputs in the Bottleneck Analysis 4.5 provide additional evidence of the current MLLMs limitations in explicit spatial perception capabilities.

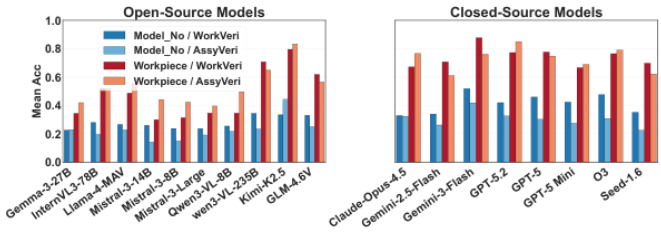

Figure 4: Performance of Open-Source vs. Closed-Source models.

D. Model-number-level tasks are more challenging for MLLMs compared to workpiece-level tasks. A comprehensive analysis of WorkVeri and AssyVeri based on Figure 4 reveals a

Table 3: Shows comprehensive benchmark results across 18 MLLMs on three manufacturing tasks, revealing significant performance gaps with best models achieving only ~70% accuracy on the most challenging tasks.

Limitations

- 3D point clouds must be converted to 2D multi-view projections due to lack of native 3D encoders in MLLMs, potentially losing spatial information
- Dataset limited to 14 workpiece categories and 90 models, may not generalize to all manufacturing domains
- Evaluation focuses on multiple-choice format which may not capture full reasoning capabilities in open-ended manufacturing scenarios
- Fine-tuning experiments only conducted on compact 3B models, scalability to larger models unclear